Telecom Customer Retention Model

Customer Churn & Customer Retention

BUSINESS PROBLEM

Business Problem

A relatively new telecom company would like to identify potential 'churners' (Customers before they leave) and take the necessary steps to retain them.

Stakeholder Audience

Primary: Executive Leadership and Marketing (Retention) Team.

Objective: Minimize customer attrition (churn) by identifying high-risk behavior patterns.

Business Logic: The cost of retention (e.g., a targeted discount) is significantly lower than the Customer Acquisition Cost. We prioritize Recall to ensure we don't miss potential leavers.

DATA

The model is trained on customer activity data, including:

- Demographics: Account length and geographic location (State/Area Code).
- Service Plans: International and Voicemail plan subscriptions.
- Usage Metrics: Daily, evening, and night charges/minutes.
- Pain Points: Frequency of customer service interactions.

Target Variable

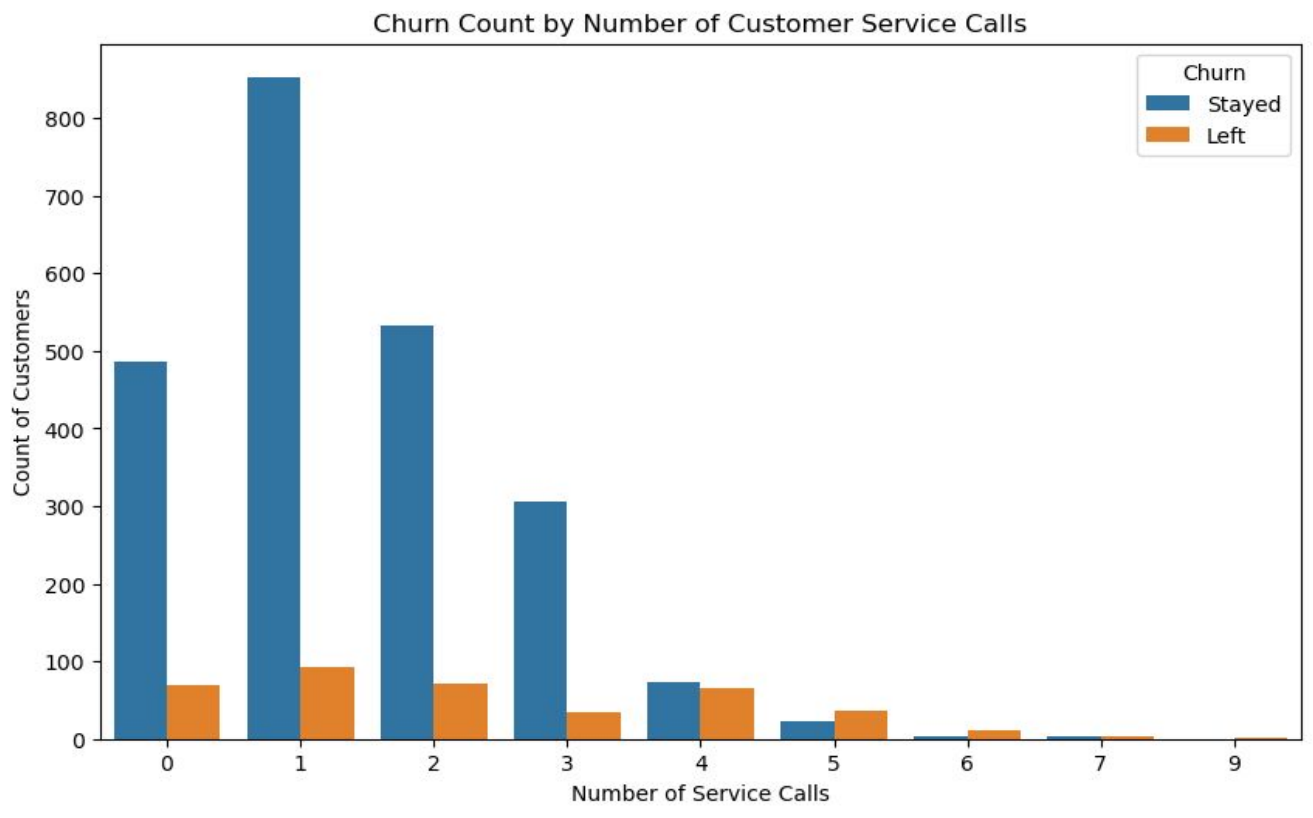
Building Models to predict Customer Churn (Customer leaving).

Class	Count	Percentage
Stayed (0)	2,850	~85.5%
Churn (Left (1))	483	~14.5%

- Majority of the customers stayed while only a small percentage left.

EXPLORATORY DATA ANALYSIS (EDA)

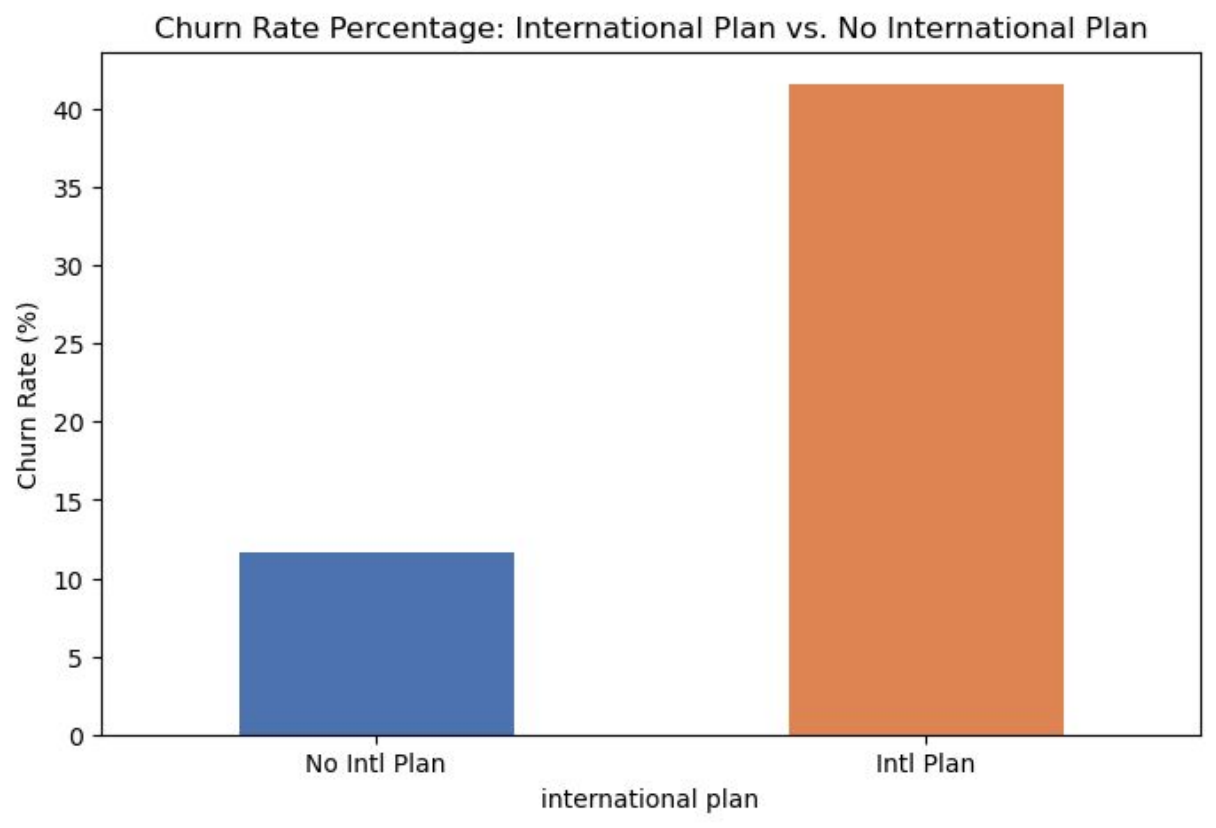
1. Exploring Service Calls and Churn



Findings:

- Customers are at high risk of leaving after 3 service calls.

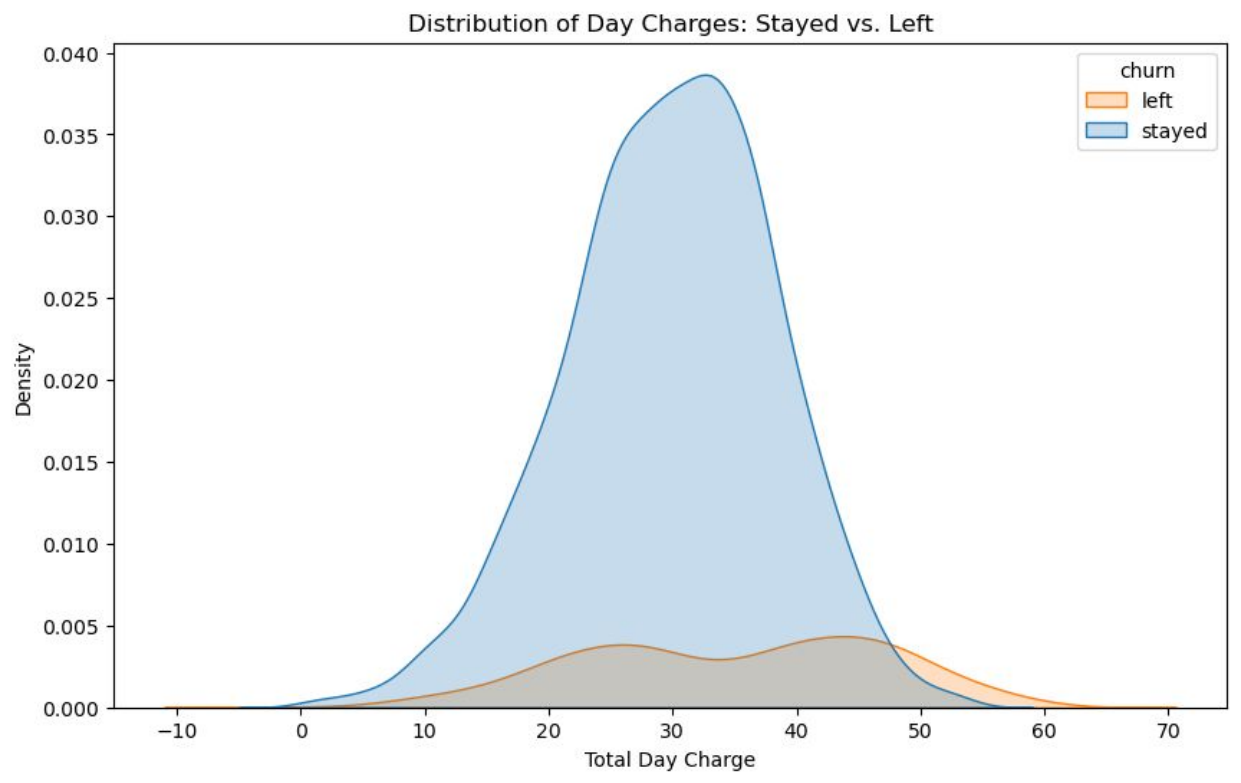
2. Exploring International Plan and Churn



Findings:

- Customers with international plans have significantly higher churn rate as compared to the ones without.

3. Exploring Total Day Charge and Churn



Findings:

- If a customer's daily charge exceeds $48, there is an overwhelming probability that they will leave.

MODELING

1
Import Data &
Basic Cleaning.

2
Train-Test Split &
EDA.

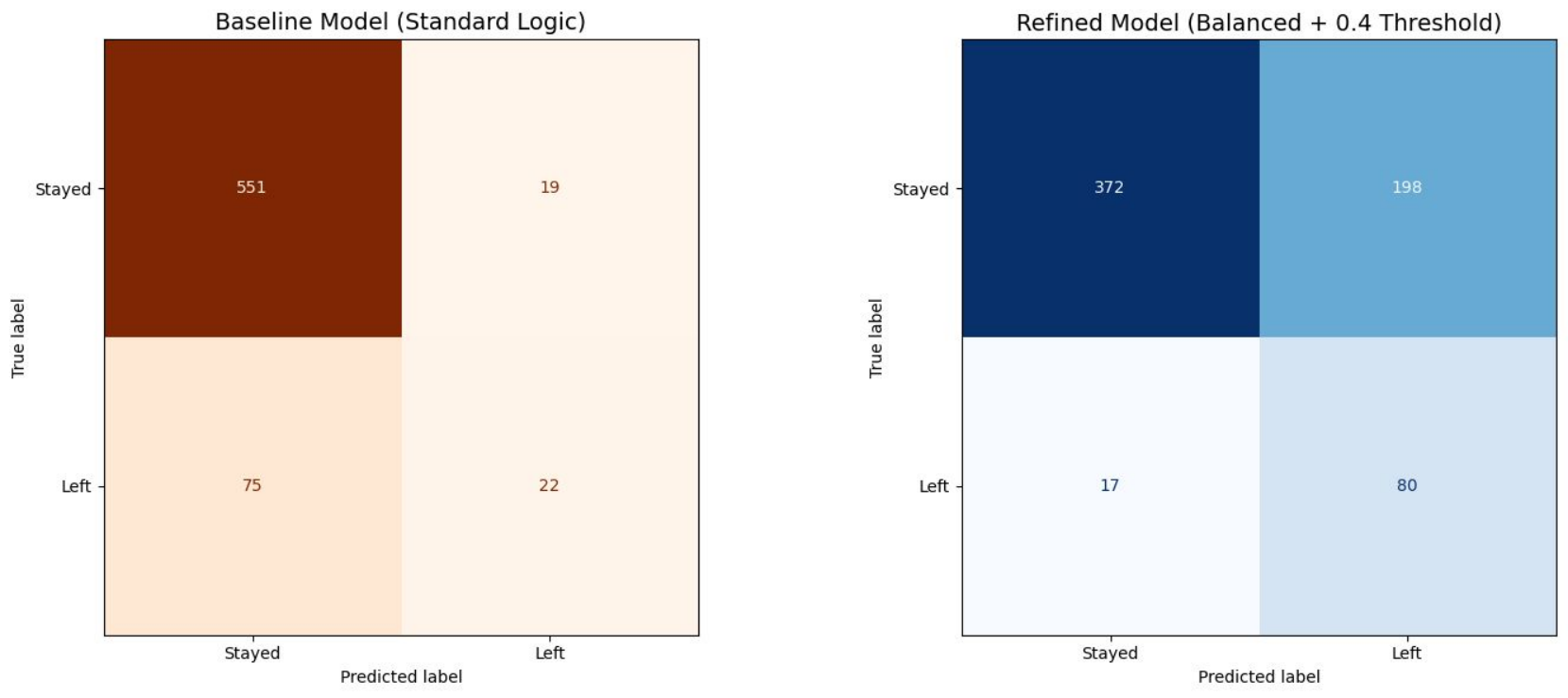
3
Preprocessing:
Data Cleaning, Standard
Scaling, Encoding

4
Modeling:
Logistic
Regression

5
Model Pruning:
Class-Weight
Balanced

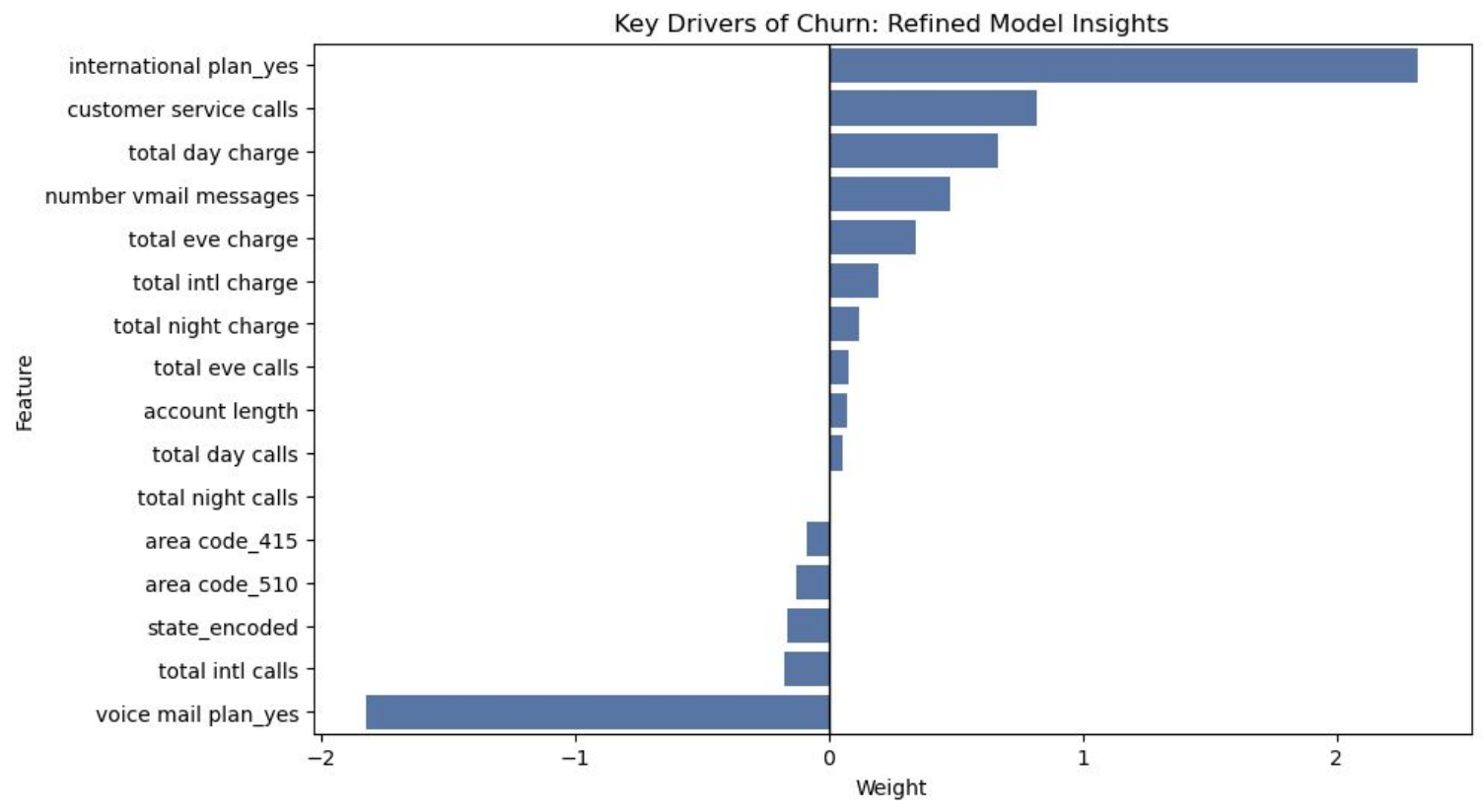
6
Model Evaluation:
Recall and
ROC-AUC score

MODEL PERFORMANCE



- The refined model (Blue) successfully captures the vast majority of churners (80) compared to the conservative baseline (Orange - 22), aligning with our "Business-First" goal of high sensitivity.

Churn Drivers from the Model



RECOMMENDATIONS

- **International Plan:** Users with this plan are the most likely to leave, suggesting a need for more competitive pricing or transparent terms.
- **Customer Service Calls:** A clear "Frustration Metric"; calls exceeding 3 per month are a major red flag.
- **Total Day Charge:** High daily spend acts as a financial trigger for churn.
- **Voicemail Plan:** Users with this plan stay loyal. New customers should be incentivised to set up voicemail.

NEXT STEPS

- **Product Analysis:** Perform a deep market review of the International Plan to identify whether customer attrition stems from pricing that isn't competitive or from technical shortcomings.
- **Operational Implementation:** Integrate the model into an active system so the Customer Service department can access up-to-the-minute churn risk insights.
- **Model Improvement:** Test advanced non-linear approaches, like Random Forest or XGBoost, to raise precision without sacrificing our 82% recall target.
- **Sentiment Monitoring:** Research on why customers are leaving after 3 customer service calls and aim at uncovering frustration indicators before clients are classified as at-risk.

THANK YOU!

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